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EDUCATION

Teesside University

Middlesbrough, UK

MSc Sustainable Engineering Systems (with Advanced Practice)

Sep. 2024 – Jun. 2026

- Coursework: Future Energy Demand, Big Data and Business Intelligence, Digital Twinning, Digital Factory Systems, Circular Economy

Teesside University

Middlesbrough, UK

BEng (Hons) Mechanical Engineering, IET and IMechE accredited

Sep. 2022 – Aug. 2024

- Dissertation: comparative evaluation of photovoltaic module encapsulant materials against thermo-mechanical durability criteria

PUBLICATION

W. Mellor, **M. Kamal**, S. M. A. Bukhari, R. Singh, G. Aggarwal, J. A. Narayanan, S. Routray, and P. M. Ushasree, “Optimisation of Silicon–Perovskite (MaPbI₃) Tandem Photovoltaic Cell With a Tunnel Junction,” *IEEE Int. Conf. on Recent Advances in Electrical, Electronics, Ubiquitous Communication and Computational Intelligence (RAEEUCCI 2026)*, Chennai, India. Scopus-indexed.
doi.org/10.1109/RAEEUCCI67649.2026.11504729

EXPERIENCE

Research Intern, Photovoltaic Device Simulation

Sep. 2025 – Jun. 2026

Teesside University, School of Computing, Engineering & Digital Technologies

Middlesbrough, UK

- Year-long placement on a joint UK–India research programme (SRM Institute of Science and Technology; SR University) modelling silicon–perovskite tandem solar cells under two academic supervisors
- Owned the silicon subcell: built and optimised the three-layer p-i-n crystalline silicon model in SCAPS-1D while a colleague developed the perovskite stack, then integrated the two
- Ran layer-thickness sweeps to lift silicon subcell efficiency from a 25.96% literature baseline to 26.03%, then added bulk defects layer by layer to locate the real losses, finding the intrinsic and n-layers far more defect-sensitive than the p-layer (18.82% and 17.13%)
- Showed that omitting defect physics inflates predicted tandem efficiency from a defensible 32.60–35.30% to an unphysical 54.78%, the paper’s central finding

PROJECTS

Lead-Free Perovskite/Silicon Tandem Solar Cell | *MSc dissertation, SCAPS-1D scripting*

2025 – 2026

- Individual project testing whether a non-toxic tin-based absorber (Cs₂SnI₆) could replace lead-based perovskites in a two-terminal tandem without losing efficiency
- Screened seven electron and hole transport layer combinations under realistic bulk and interface defects, finding an NiO/SnO₂ pairing best at 32.74% efficiency against 26.87% for the organic PEDOT:PSS/C₆₀ baseline
- Wrote a SCAPS-1D script that filters the solar spectrum through the top cell, feeds the transmitted light to the silicon subcell, and sweeps absorber thickness in 5 nm steps to find the current-matched geometry at 505 nm

Sales Analytics Dashboard | *Power BI, DAX*

2025

- Four-page interactive report analysing revenue, profit and order volume by product category, country and customer demographic
- Wrote DAX measures for revenue, profit, average order value, markup percentage and revenue per customer, with a disconnected year table driving dynamic period comparison across every visual

Digital Factory Case Study | *Siemens Tecnomatix Plant Simulation*

2025

- Individual 4,000-word case study on deploying a digital factory for a specific engineering application, weighing throughput gains against security, safety and risk constraints, supported by production-line modelling in Plant Simulation

TECHNICAL SKILLS

Software: SCAPS-1D (device physics, scripting), Power BI (DAX, data modelling), Siemens Tecnomatix Plant Simulation, Siemens NX, Microsoft Excel

Domain: Photovoltaic device performance, energy demand, digital twins, life-cycle assessment, Industry 4.0

Other: Technical and academic writing, literature review, international research collaboration. English and Arabic.